Linyi (Tiger) Hou

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Education

Ph.D. in Aerospace Engineering, University of Illinois Urbana-Champaign — Jun 2021 - present
B.S. in AE, Minor in CompE, University of Illinois Urbana-Champaign — Aug 2017 - May 2021

Highlights for ADCS Position

UAV Navigation and Control

Teaching Assistant — 2021-2023

- Implemented an LQR controller and observer in firmware using C with FreeRTOS.
- Helped students formulate and implement state-space model controllers with various sensor configurations, including gyroscopes, laser rangefinders, optical flow sensors, and time-of-flight anchors.

Space Grant Midwest High-Power Rocketry Competition

Avionics Lead — 2017-2018

- Led team of 3 to design and manufacture a flap-based roll control system by fusing gyroscope and magnetometer data.

Work Experience

University of Illinois Urbana-Champaign

Teaching Assistant (AE 202, AE 443, AE 483) — Jun 2021 - present

- Worked with 2-5 faculty and staff to teach lessons, hold office hours, and improve curriculum for courses in flight mechanics, systems engineering, and UAV controls.
- Assisted in developing code and instructions for a UAV lab based on C and Python.

Collins Aerospace

Software Engineering Intern — May 2019 - Aug 2019

- Migrated an avionics data interpreter from 16-bit to 32-bit using C++ and Python.
- Automated builds in TortoiseSVN using Python, reducing need for manual integration.

Research

Astrodynamics and Planetary Exploration Group (APEX) — Advisor: Dr. Siegfried Eggl

Interstellar Navigation and Star Identification — Sep 2024 - present

- Developing star identification techniques for interstellar travel at relativistic speeds by combining traditional algorithms with clustering based on stellar classification.

Celestial Navigation to Supplement DSN Tracking Jul 2024 - present

- Collaborating with four colleagues to assess feasibility of using celestial navigation to meet cruise phase uncertainty requirements in heliocentric and cislunar space.
- Designing sensor tasking/scheduling algorithms with novel considerations for covariance propagation and visibility constraints over the time domain.
- Implemented spherical harmonics for an open source orbit propagator in C++.

Lost in Space and Time Navigation using Variable Stars Nov 2022 - present

- Developed an autonomous optical navigation technique based on light curve analysis of variable stars which does not require prior knowledge of position or time.
- Simulated star tracker observations and performed Monte Carlo analysis to assess the feasibility of reestablishing Earth pointing via the proposed method.

Aerospace Mission Analysis Laboratory Advisor: Dr. Zachary R. Putnam

Lost in Space X-ray Pulsar Navigation (XNAV) Jun 2021 - present

- Developed algorithms to resolve spacecraft position and time from pulsar timing data.
- Created a photon time of arrival simulation tool in Python and validated results against the pulsar timing package PINT to within 50 µs.

Velocity-based Initial Orbit Determination Aug 2019 - Apr 2024

- Developed a novel initial orbit determination (IOD) technique using sequential range-rate measurements to distant stars.
- Assessed the performance of velocity IOD methods using Monte Carlo simulation.

Skills

Programming Languages	Python, C/C++, MATLAB/Simulink, Javascript
Software/Prepackaged Tools	LaTeX, Git, GMAT
Operating Systems	Linux, MacOS, Windows
Languages	English, Mandarin Chinese

Honors and Awards

Yee Fellowship 2021, 2024
Grainger College of Engineering, University of Illinois Urbana-Champaign

Aerospace Engineering Alumni Advisory Board Fellowship 2024
Department of Aerospace Engineering, University of Illinois Urbana-Champaign

Best Student Paper, 46$^{\text{th}}$ AAS GN&C Conference 2024
American Astronautical Society (AAS) Rocky Mountain Section

David and Catherine Thompson Space Technology Scholarship 2020
American Institute of Aeronautics and Astronautics (AIAA)